

RAHUL DEBNATH

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Professional Summary

AI/ML Intern with hands-on production experience building Agentic AI pipelines, RAG systems, LLM-powered applications, and Computer Vision models across enterprise and international client projects. At RMSI, architected PolicyMind AI — a multi-agent, evidence-grounded document intelligence system — and delivered a Geospatial AI product for the City of Calgary achieving 95.15 % accuracy classifying urban tree species from 490 GB of aerial imagery. Experienced across the complete ML lifecycle: data engineering, deep learning, model evaluation, MLOps, and production deployment. Work spans Generative AI, multimodal systems, and remote sensing research. IEEE IGARSS 2026 accepted author. Core stack: Python, PyTorch, FastAPI.

Professional Experience

RMSI Pvt. Ltd.

AI/ML Intern

Jan 2026 – Present

Noida, Uttar Pradesh, India

- Own end-to-end **production AI/ML** delivery for international clients across the full ML lifecycle.
- Engineer and deploy **Agentic AI**, **RAG**, **LLM**, and **Computer Vision** systems in a professional production environment.
- Deliver enterprise **Geospatial AI** and **Remote Sensing** solutions across urban analytics and environmental monitoring.

Adamas University

Geospatial Data Science Researcher

Jun 2025 – Jul 2025

Kolkata, India

- Conducted **ML**-driven spatial analysis on large **remote sensing** datasets; contributed to a peer-reviewed publication.
- Processed multi-source geospatial data using **Python**, **ArcGIS**, and **QGIS** for statistical modeling and visualization.

Skills

- **Programming & Databases:** Python, SQL, PostgreSQL, NumPy, Pandas, Scikit-learn, Gdal
- **AI & Machine Learning:** Agentic AI, Large Language Models (LLMs), RAG, Generative AI, Prompt Engineering, Transformers, Decision Trees, Random Forest, Linear/Logistic Regression, SVM
- **Deep Learning & Frameworks:** CNN, ANN, RNN, TensorFlow, PyTorch, Transfer Learning, HuggingFace
- **Computer Vision:** Image Classification, Object Detection, Image Segmentation, Multimodal AI, Vision–Language Models, OpenCV
- **NLP & Information Retrieval:** Text Mining, Tokenization, NER, Sentiment Analysis, Semantic Search, Vector Databases (FAISS, pgvector, Pinecone)
- **MLOps & Deployment:** FastAPI, Gradio, Streamlit, Groq API, Production Model Deployment
- **Geospatial AI & Remote Sensing:** GIS, SAR Analysis, Change Detection, Spatial Data Analytics, Geostatistics, Google Earth Engine (GEE), GDAL, Rasterio, Fiona, GeoPandas, Shapely, QGIS, SNAP
- **Statistics & Analytics:** Hypothesis Testing, Regression Analysis, Feature Engineering, Time Series Forecasting
- **Cloud & DevOps:** Google Cloud Platform (GCP), Git, GitHub
- **Data Visualization:** PowerBI, Matplotlib, Seaborn

Education

M.Sc. Data Analytics with Specialization in Geoinformatics

Kerala University of Digital Sciences, Innovation and Technology (Formerly IIITM-K)

Aug 2024 – Present

Thiruvananthapuram, Kerala, India

– Semester Scores: Sem 1 – 7.15 | Sem 2 – 8.85 | Sem 3 – 9.25

Bachelor of Science in Geography (Honours)

Ranaghat College, Affiliated with Kalyani University

2021 – 2024

Ranaghat, West Bengal, India

– CGPA: 7.51 / 10

Projects

PolicyMind-AI — Agentic RAG Insurance Document Intelligence

May 2026

- *Production-deployed product at RMSI Private Limited* for extracting structured insurance data from heterogeneous policy PDFs into a standardized JSON schema.
- Engineered an end-to-end **Agentic AI and RAG pipeline** combining OCR, page classification, deterministic extraction, semantic retrieval, specialist LLM agents, evidence validation, confidence scoring, and human-correction memory.
- Developed production **FastAPI** services and an evidence-grounded PDF chatbot using **Gemini, Groq, FAISS/pgvector, Hugging Face embeddings, Python, and PostgreSQL**.
- **Skills:** Agentic AI, RAG, LLMs, Prompt Engineering, OCR, NLP, Vector Search, FastAPI, FAISS, PostgreSQL, Python

Urban Tree Species Classification System: Geospatial AI Product for Calgary

Jan 2026

- Developed an end-to-end **geospatial DL pipeline** to classify **9 urban tree species** from high-resolution **0.1 m RGB aerial orthomosaic imagery**, supporting automated urban tree inventory management at city scale.
- Trained and evaluated deep learning models using **PyTorch, Torchvision, and ResNet18 ImageNet transfer learning**, achieving **95.15% validation accuracy**
- Engineered large-scale remote sensing workflows for **490 GB GeoTIFF imagery**, including **MrSID-to-GeoTIFF conversion**, image patch extraction, class-balanced dataset creation, geospatial quality control, and VRT-based raster optimization.
- Deployed the trained model as a **FastAPI REST service** with database-driven inference, background job tracking, error logging, and integration readiness for enterprise AI/ML APIs.
- **Skills:** Machine Learning, Deep Learning, Computer Vision, Remote Sensing, Geospatial AI, Transfer Learning, PyTorch, Torchvision, ResNet18, FastAPI, Rasterio, GeoPandas, GDAL, QGIS, OpenCV, scikit-learn, GeoTIFF, VRT, MLOps

Med-Echo: Multimodal AI-Powered Medical Voice Assistant

Nov 2025

- *Generative AI Course Project, Digital University Kerala* [in LinkedIn](#) — [GitHub](#)
- Developed an **AI-powered medical consultation system** integrating **Whisper V3** (speech recognition), **Llama 4 Scout** (vision-language model), and **neural TTS** for real-time multimodal clinical query interpretation.
- Built a **Gradio web interface** for voice input and image upload, achieving **<20s end-to-end latency**
- **Skills:** LLMs, VLMs, Speech Recognition, TTS, Python, Groq API, Gradio, Multimodal AI

Multi-Temporal Detection of Illegal Iron Ore Mining — Jharkhand & Odisha

Oct 2025

- *Mini Project (3rd Semester), Digital University Kerala*
- [\[IGARSS 2026 ACCEPTED — Manuscript #2043\]](#) Research paper titled "*Multi-Temporal Detection of Small-Scale Iron Ore Mining Areas Using Integrated SAR and Optical Data with Machine Learning*" **accepted for oral/poster presentation at IEEE IGARSS 2026** by the Technical Program Committee — one of the most prestigious international remote sensing conferences.
- Developed a machine learning-based change detection workflow using multi-temporal **Sentinel-1 (SAR)** and **Sentinel-2 (optical)** data, integrated with spectral indices (NDVI, NDBI, BSI) in **Google Earth Engine** and **QGIS** for illegal mining area identification and spatial visualization.
- **Skills:** Remote Sensing, ML, SAR, Sentinel-1/2, GEE, Python, QGIS

Generative AI for Synthetic Soil Nutrient Map Generation

Sep 2025

- *Spatial Data Analytics Course Project, Digital University Kerala* [GitHub](#)
- [\[IEEE MIGARS 2026 ACCEPTED — Manuscript #328\]](#) Research paper titled "*AI-Driven Soil Nutrient Mapping Using Geostatistical, Machine Learning, and Deep Learning Approaches*" **accepted for poster presentation at IEEE IGARSS 2026** by the Technical Program Committee.
- Designed a **Diffusion Model-based Generative AI system** to simulate and impute missing soil nutrient data, integrating **remote sensing datasets** and **SoilGrids** for enhanced spatial prediction in sustainable agriculture.
- **Skills:** Generative AI, Diffusion Models, GeoAI, Python, ML, Remote Sensing

Additional Information

- **Languages:** English (Professional), Hindi (Fluent), Bengali (Native)
- **Research Interests:** AI/ML, GeoAI, Satellite Image Analysis, Multimodal AI, Environmental Remote Sensing, Deep Learning for Earth Observation